

A mathematical modelling of rheumatoid arthritis

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Abstract

We developed a mathematical model to describe the inflammatory dynamics during arthritis development and resolution, focusing on the interaction between three key immune cell populations within the joint, namely neutrophils (N), synovial sublining tissue-resident macrophages (SL-TRM, T), and circulating monocytes that are recruited into the inflamed joint and differentiate into monocyte-derived macrophages (MoMac, M). An inflammatory signal (S), representing the response to an external arthritogenic stimulus, drives the recruitment and activation of neutrophils and MoMac, while SL-TRM exert a regulatory and protective role. We propose a minimal ordinary differential equation model of acute inflammatory response into a joint that couples an inflammatory signal S to the three cell populations (N, T, M) and incorporates recruitment, conversion, nonlinear self-limitation and mutual regulation. Using nondimensionalization, we showed that the equilibrium structure depends on three adimensional parameters A, B, C and derives a cubic equation characterizing nontrivial steady states. We performed an analytical study of equilibria and their stability, and identified a bifurcation at $A = 1$ separating a disease-free regime from a sustained inflammatory state, and discussed conditions for multiple biological feasible steady states, consistent with transient or persistent inflammation. Numerical simulations illustrate typical transient dynamics and phase- space geometry. In conclusion, our model provides a framework to explore mechanisms underlying acute versus sustained joint inflammation and offers a basis for parameter estimation from experimental data, and future extensions toward evaluation of therapeutic strategies.

1 Model

In the experimental mouse model of arthritis used in this study, the acute inflammatory response is triggered by an external signal (injection of an arthritogenic serum) that activates the immune system and initiates an inflammatory response within the joint tissue, which is composed of a lining and sub-lining membrane. At steady state, the synovial lining membrane is populated by specialized tissue-resident macrophages (L-TRM) that

form a protective barrier and contribute to immune regulation. TRM are also present within the sublining membrane (SL-TRM), scattered and fewer in number. Both TRM contribute to early immune sensing of environmental cues, exerting an overall protective and regulatory role by limiting excessive inflammation. However, during arthritis inflammatory flares, while L-TRM remain preserved, SL-TRM are progressively depleted. This depletion is concomitant with the accumulation of macrophages derived from blood monocytes infiltrating the inflamed joints (MoMac). The loss of the SL-TRM population weakens synovial immune regulation and favors persistent inflammatory cell infiltration through to the local production of cytokines and chemokines. This activation promotes the rapid recruitment of neutrophils (N) from the bloodstream into the joint, where they become the dominant cell population during the acute phase of inflammation. Neutrophils amplify the inflammatory response through the release of pro-inflammatory mediators. Based on these biological observations, we modeled the joint inflammatory response as the interaction between an inflammatory signal and three cell populations—N, SL-TRM, and MoMac—capturing recruitment, amplification, and regulatory mechanisms in a minimal dynamical framework.

1.1 Mathematical Model

Let us denote by $T(t)$ the population of SL-TRM cells and by $M(t)$ the population of Momac cells at time t . The cells T (SL-TRM) and M (Momac) interact with each other in the following way. The Momac cells M are recruited from the blood stream at a constant rate a . They die at a rate δ_M and they transform into SL-TRM cells at a rate b . The SL-TRM cells T die at a rate δ_T . The interaction between the two populations is described by the following system of differential equations

The inflammatory state $S(t)$ is enhanced by neutrophils N and inhibited by SL-TRM macrophages T .

Inflammation further increases neutrophils (factor α), increases Momac (factor β), and decreases SL-TRM (factor γ).

Notice that we do not keep track of the population of fibroblasts, but rather the population of neutrophils, which are recruited by inflammatory chemokines.

The variable $N(t)$ designates the population of neutrophils at time t . Finally $S(t)$ is the inflammatory signal at time t . The model is given by the following system of ordinary differential equations:

$$\left\{ \begin{array}{l} \frac{dS}{dt} = (a N - b T)S \\ \frac{dN}{dt} = \alpha S N - \delta_N N(N - 1) \\ \frac{dM}{dt} = \beta S - \delta_M M(M - 1) \\ \frac{dT}{dt} = -\gamma S T + \delta_M M(1 - T) \end{array} \right. \quad (1)$$

where $a, b, \alpha, \beta, \gamma, \delta_N, \delta_M$ are positive parameters. To make things easier to understand, we summarize the model in the following diagram (see Figure 1.1).

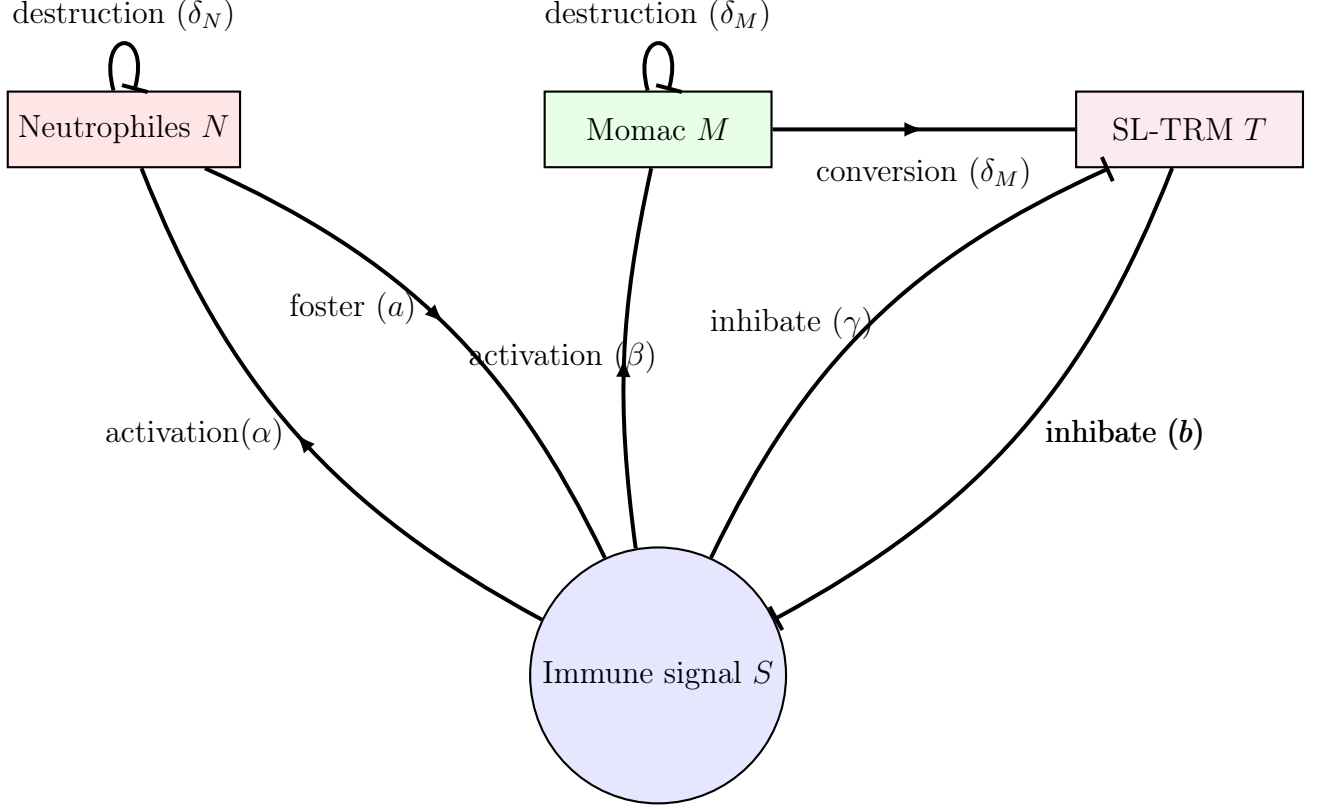


Figure 1: Schematic representation of the model

2 Analysis of the model

2.1 Existence and non-negativity of the solution.

Existence and uniqueness of a local solution of the system (1) is guaranteed by the Cauchy-Lipschitz theorem since the right-hand side of the system is locally Lipschitz continuous.

Proposition 2.1. *If the initial conditions $S(0)$, $N(0)$, $M(0)$, and $T(0)$ are nonnegative, then the solution $(S(t), N(t), M(t), T(t))$ of the system (1) remains nonnegative for all $t \geq 0$.*

Proof. We consider the vector field $F = (f_1, f_2, f_3, f_4)$ for $y = (y_1, y_2, y_3, y_4) \in \mathbb{R}^4$ given by:

$$\begin{cases} f_1(y) = (ay_2 - by_4)y_1, \\ f_2(y) = \alpha y_1 y_2 - \delta_N y_2 (y_2 - 1), \\ f_3(y) = \beta y_1 - \delta_M y_3 (y_3 - 1), \\ f_4(y) = -\gamma y_1 y_4 + \delta_M y_3 (1 - y_4). \end{cases}$$

We observe that the vector field F is *quasi-positive*, i.e. for each $i = 1, 2, 3, 4$, we have $f_i(y) \geq 0$ whenever $y_i = 0$ and $y_j \geq 0$ for all $j \neq i$. Thus from Proposition 2.1 in [3], we deduce that if the initial conditions are nonnegative, then the solution remains nonnegative for all $t \geq 0$. $\square$

This property is biologically meaningful since the variables S , N , M , and T represent cell populations, which cannot be negative. Our model thus ensures that the solutions remain within the biologically relevant domain.

Remark 2.1. *We did not include degradation terms in the S, N, M equations, which could ensure the boundedness of the solution. Indeed we shall see that in some cases the solution may not be bounded for all time, at least for S, N and M . Actually we are interested in the immune response to an external stimulus, which can be very strong and can lead to a very high level of inflammation and cell recruitment from the bloodstream or marrow. The transient dynamics of the system can be very complex and can involve large variations in the levels of inflammation and cell populations, which may not be captured by a model that includes degradation terms that dampens the growth of these variables. In this case, the solution may not be bounded for all time, but it can still be biologically relevant since we are interested in the crisis dynamics rather than the long-term behavior of the system. In future work, we could consider adding degradation terms to the equations to ensure boundedness of the solution, but this would also add complexity to the model and may not be necessary for capturing the transient dynamics of interest. However the solution is always bounded in the T variable, since $T(t) \leq 1$ for all $t \geq 0$. This is biologically relevant since the population of SL-TRM cells cannot exceed a certain maximum level due to spatial constraints within the joint tissue and regulatory mechanisms that limit their proliferation.*

2.2 Equilibria points

Equilibria points of the system are found by setting the right-hand side of the system to zero, i.e. by solving the following system of equations: The equilibrium points (S^*, N^*, M^*, T^*) satisfy the following system:

$$(aN^* - bT^*)S^* = 0 \quad (2)$$

$$\alpha S^* N^* - \delta_N N^* (N^* - 1) = 0 \quad (3)$$

$$\beta S^* - \delta_M M^* (M^* - 1) = 0 \quad (4)$$

$$-\gamma S^* T^* + \delta_M M^* (1 - T^*) = 0 \quad (5)$$

2.2.1 Inflammation free equilibria ($S^* = 0$).

A first natural equilibrium point is given by $S^* = 0, N^* = 1, M^* = 1, T^* = 1$. This point corresponds to the disease-free state, where there is no inflammation ($S^* = 0$) and the populations of neutrophils, Momac, and SL-TRM are at their baseline levels ($N^* = 1, M^* = 1, T^* = 1$).

There are other equilibria points with $S^* = 0$ namely $S^* = 0, N^* = 0, M^* = 0, T^* = R$, with arbitrary R ; $S^* = 0, N^* = 0, M^* = 1, T^* = 1$; $S^* = 0, N^* = 1, M^* = 0, T^* = R$, with arbitrary R .

Remark 2.2. *These equilibria points with $S^* = 0$ and $N^* = 0$ or $M^* = 0$ are not biologically relevant since they correspond to the absence of neutrophils or Momac cells, which is not realistic in a living organism. Anyway, we will prove later that these equilibria points are unstable, so they will not be observed in practice.*

2.2.2 Equilibria points with $S^* \neq 0$.

We express N^* in terms of T^* from equation (2):

$$N^* = \frac{b}{a}T^* \quad (6)$$

Substituting this expression for N^* into equation (3), we obtain:

$$S^* = \frac{\delta_N}{\alpha} \left(\frac{b}{a}T^* - 1 \right) \quad (7)$$

From equation (4) and equation (5), we derive $\delta_M M^* = \frac{\beta S^*}{M^* - 1}$ and $\delta_M M^* = \frac{\gamma S^* T^*}{1 - T^*}$. Equating these two expressions for $\delta_M M^*$, we obtain:

$$\frac{\beta S^*}{M^* - 1} = \frac{\gamma S^* T^*}{1 - T^*}$$

Assuming $S^* \neq 0$, we can simplify this equation to:

$$\frac{\beta}{M^* - 1} = \frac{\gamma T^*}{1 - T^*}$$

From this, we can express M^* in terms of T^* :

$$M^* = 1 + \frac{\beta(1 - T^*)}{\gamma T^*} = 1 - \frac{\beta}{\gamma} + \frac{\beta}{\gamma T^*} \quad (8)$$

Substituting the expression for M^* and S^* into the last equation (5), we obtain

$$-\gamma \frac{\delta_N}{\alpha} \left(\frac{b}{a}T^* - 1 \right) T^* + \delta_M \left(1 - \frac{\beta}{\gamma} + \frac{\beta}{\gamma T^*} \right) (1 - T^*) = 0 \quad (9)$$

Since $T^* \neq 0$, we can multiply this equation by T^* and rearrange the terms to obtain a cubic equation for T^* :

$$\frac{\gamma}{\alpha} \frac{\delta_N}{\delta_M} \frac{b}{a} (T^*)^3 + \left(1 - \frac{\gamma}{\alpha} \frac{\delta_N}{\delta_M} - \frac{\beta}{\gamma} \right) (T^*)^2 + \left(2\frac{\beta}{\gamma} - 1 \right) T^* - \frac{\beta}{\gamma} = 0 \quad (10)$$

Rewriting this equation in the standard unitary form $(T^*)^3 + p(T^*)^2 + qT^* + r = 0$, we have:

$$(T^*)^3 + \frac{\alpha}{\gamma} \frac{\delta_M}{\delta_N} \frac{a}{b} \left(1 - \frac{\beta}{\gamma} - \frac{\gamma}{\alpha} \frac{\delta_N}{\delta_M} \right) (T^*)^2 + \frac{\alpha}{\gamma} \frac{\delta_M}{\delta_N} \frac{a}{b} \left(2\frac{\beta}{\gamma} - 1 \right) T^* - \frac{\alpha}{\gamma} \frac{\delta_M}{\delta_N} \frac{a}{b} \frac{\beta}{\gamma} = 0 \quad (11)$$

Notice that we need only 3 *non dimensional* parameters a/b , α/β , $\frac{\gamma\delta_N}{\alpha\delta_M}$ instead of the 7 original parameters. In the following, we denote the 3 *non dimensional* parameters as $A = a/b$, $B = \beta/\gamma$, and $C = \frac{\gamma\delta_N}{\alpha\delta_M}$.

Remark 2.3. *The parameters A , B , and C have the following biological interpretations:*

- $A = a/b$. *This parameter represents the ratio of the inflammatory effect of the neutrophils w.r.t. the anti-inflammatory effect of the macrophages.*

- $B = \beta/\gamma$. This parameter represents the ratio of the increasing rate of the Momac to the decreasing rate of the macrophage.
- $C = \frac{\gamma \delta_N}{\alpha \delta_M} = \left[\frac{\gamma}{\delta_M} : \frac{\alpha}{\delta_N} \right]$. This parameter represents the ratio of the macrophages vs the neutrophils effects in the disease progression.

Using these parameters, the cubic equation in T^* becomes:

$$(T^*)^3 + \frac{A}{C}(1 - B - C)(T^*)^2 + \frac{A}{C}(2B - 1)T^* - \frac{AB}{C} = 0 \quad (12)$$

Remark 2.4. There is an obvious solution in the particular case $A = a/b = 1$, namely $T^* = 1$. This solution corresponds to the disease-free equilibrium point $S^* = 0, N^* = 1, M^* = 1, T^* = 1$. But this solution is not relevant for the case $S^* \neq 0$ that we are considering here. Note further that if $T^* = 1$ then from Eq. (12) we get

$$1 + \frac{A}{C}(1 - B - C) + \frac{A}{C}(2B - 1) - \frac{AB}{C} = 0$$

so that $A = 1$ hence $T = 1$ is a root of the cubic equation (12) iff $A = 1$. Therefore we can factor the cubic polynomial as follows:

$$(T^* - 1) \left((T^*)^2 - \frac{1}{C}(B - 1)T^* + \frac{B}{C} \right) = 0 \quad (13)$$

- If $B \leq 1$, then the coefficients of the quadratic factor are positive, hence there are no other positive roots.
- If $B > 1$, the quadratic factor has either two positive roots (if the discriminant is positive) or no real roots (if the discriminant is negative). The quadratic factor has discriminant $\Delta = \frac{(B-1)^2 - 4BC}{C^2}$. Thus, we have:

- If $\Delta > 0$, there are two distinct real roots given by:

$$T^* = \frac{\frac{(B-1)}{C} \pm \sqrt{\Delta}}{2}$$

- If $\Delta = 0$, there is one (double) positive root given by:

$$T^* = \frac{B - 1}{2C}$$

- If $\Delta < 0$, there are no real roots.

The sign of the discriminant of the quadratic factor depends on the parameters B and C : $\Delta > 0$ iff $\frac{(B-1)^2}{4B} > C$. Let us plot the curve $\frac{(B-1)^2}{4B}$ in the (B, C) plane (see Figure 2). Looking at the figure 2, we see that the typical situation is when $\Delta < 0$, hence there are no other positive roots apart from $T^* = 1$. But for a careful choice of parameters B and C , we can indeed have three equilibria, see fig. 3.

Returning to the general case $A \neq 1$, we analyze the number of positive roots of the cubic equation (12). From basic algebra, we can show that there are at most 3 such intersection points, hence at most 3 equilibria points with $S^* \neq 0$. We can be more precise because the biological feasibility imposes the roots to be real nonnegative numbers.

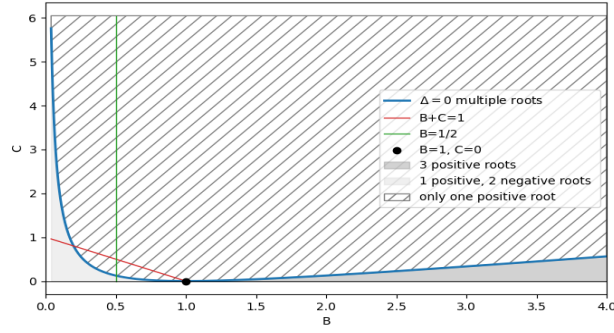


Figure 2: $A = 1$, number of roots.

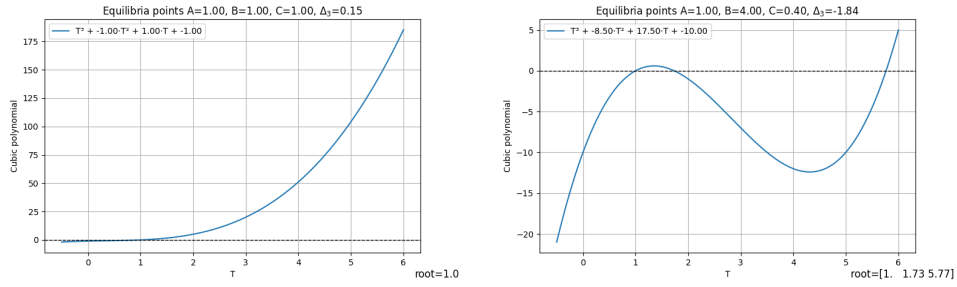


Figure 3: $A = 1$, Left 1 positive root, Right 3 positive roots

Since $\frac{AB}{C} > 0$, there is at least one positive root. But we can be more precise using *Descartes' rule of signs*. It states that the number of positive real roots of a polynomial (counting multiplicities) is equal to the number of sign changes between consecutive nonzero coefficients, or less than that by an even number. In our case, the coefficients of the cubic polynomial in (12) are:

$$1, \quad \frac{A}{C}(1 - B - C), \quad \frac{A}{C}(2B - 1), \quad -\frac{AB}{C}$$

So the number of sign changes depends on the signs of the second and third coefficients. We have the following cases:

- If $B + C > 1$ and $B > 1/2$, then there are 3 sign changes and the cubic has one or three positive roots, which can collapse according to their multiplicities.
- If $B + C < 1$ and $B > 1/2$, then there is one sign change and the cubic has only one positive roots.
- If $B + C < 1$ and $B < 1/2$, then there is one sign changes and the cubic has only one positive root.
- If $B + C > 1$ and $B < 1/2$, then there is one sign change and the cubic has only one positive root.

We can plot some typical cases in the following figure (see Figure 4).

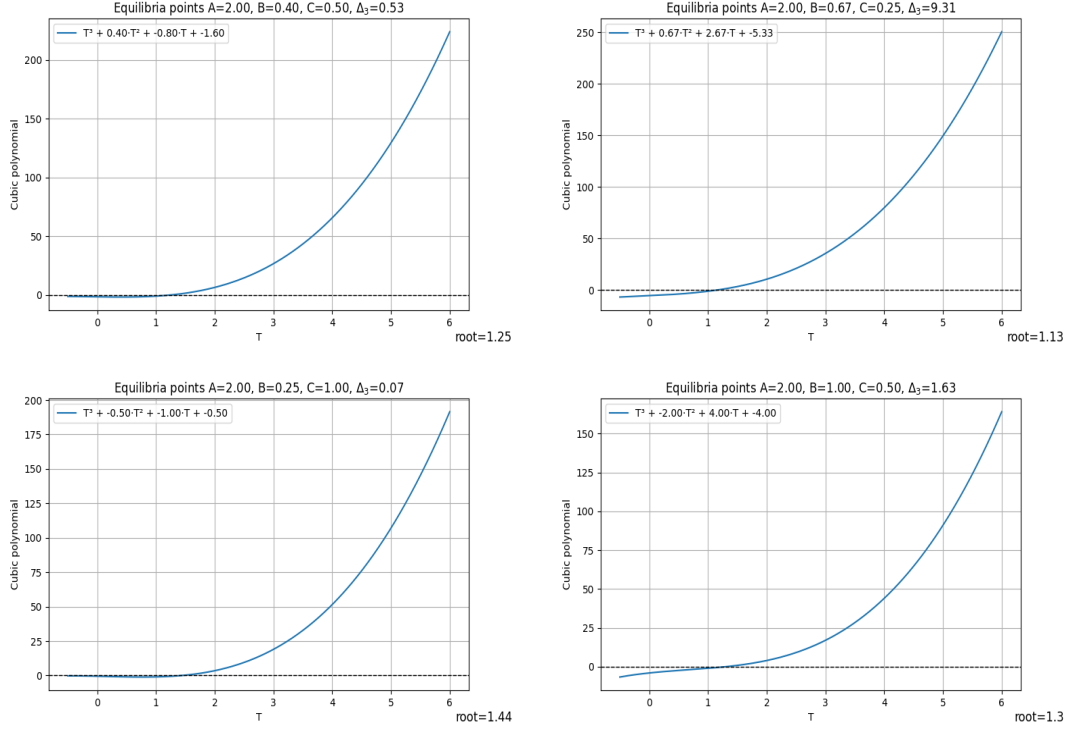


Figure 4: Cubic equation in T^* , left $B < 1/2$, right $B > 1/2$.

We can use Cardano's method to find the roots of the cubic equation (12). We rewrite it in the standard form:

$$T^3 + pT^2 + qT + r = 0$$

where

$$p = \frac{A}{C}(1 - B - C), \quad q = \frac{A}{C}(2B - 1), \quad r = -\frac{AB}{C}$$

We perform the change of variable $T = x - p/3$ to eliminate the quadratic term and obtain the depressed cubic equation:

$$x^3 + \mathcal{A}x + \mathcal{B} = 0$$

where

$$\mathcal{A} = q - \frac{p^2}{3}, \quad \mathcal{B} = \frac{2p^3}{27} - \frac{pq}{3} + r$$

The discriminant of this equation is given by:

$$\Delta_3 = \left(\frac{\mathcal{B}}{2}\right)^2 + \left(\frac{\mathcal{A}}{3}\right)^3$$

The nature of the roots depends on the sign of the discriminant Δ :

- If $\Delta_3 > 0$, there is one real root and two complex conjugate roots.
- If $\Delta_3 = 0$, all roots are real and at least two are equal.
- If $\Delta_3 < 0$, all roots are real and unequal.

Hence we can characterize the case when we have three distinct real roots by the condition $\Delta_3 < 0$, which is equivalent to:

$$\left(\frac{\mathcal{A}}{3}\right)^3 < -\left(\frac{\mathcal{B}}{2}\right)^2$$

This condition can be rewritten in terms of the original parameters using computer algebra as follows:

$$\left(\frac{\mathcal{A}}{3}\right)^3 < -\left(\frac{\mathcal{B}}{2}\right)^2$$

where

$$\begin{aligned}\mathcal{A} &= \frac{A}{3C^2} \left(-A(B+C-1)^2 + 3C(2B-1) \right) \\ \mathcal{B} &= \frac{A}{27C^3} \left(-2A^2(B+C-1)^3 + 9AC(2B-1)(B+C-1) - 27BC^2 \right) \\ \Delta_3 &= \frac{A^2}{108C^4} \left(A^2(4BC-1)(B+C-1)^2 - 2AC(2B-1)(B^2-9BC-B-2) + 27B^2C^2 \right)\end{aligned}$$

The real roots can be found using Cardano's formulas. If $\Delta_3 > 0$, the real root is given by:

$$x^* = \sqrt[3]{-\frac{\mathcal{B}}{2} + \sqrt{\Delta_3}} + \sqrt[3]{-\frac{\mathcal{B}}{2} - \sqrt{\Delta_3}}$$

If $\Delta_3 < 0$, the three real roots are given by:

$$x_k^* = 2\sqrt{-\frac{\mathcal{A}}{3}} \cos \left(\frac{1}{3} \arccos \left(\frac{3\mathcal{B}}{2\mathcal{A}} \sqrt{-\frac{3}{\mathcal{A}}} \right) + \frac{2k\pi}{3} \right), \quad k = 0, 1, 2$$

If $\Delta_3 = 0$, the real roots are given by:

$$x_1^* = 2\sqrt[3]{-\frac{\mathcal{B}}{2}}, \quad x_2^* = x_3^* = \sqrt[3]{\frac{\mathcal{B}}{2}}$$

Here we plot a case with three distinct positive roots (see Figure 5). Namely we take $A = 1.1$, $B = 5$ and $C = \frac{2}{3}$.

Finally, we can recover the roots T^* of the original cubic equation using the relation $T^* = x^* - p/3$.

We have to make sure that the nonnegative roots T^* found are biologically feasible, i.e. that they lead to nonnegative values of S^* , N^* , and M^* . For instance in the case $A = 1.1$, $B = 5$, $C = 2/3$ plotted in Figure 5, we have three distinct positive roots of the cubic equation (12), but none satisfying the biological feasibility conditions.

From equations (6), (7), and (8), we have:

$$N^* = \frac{b}{a}T^*, \quad S^* = \frac{\delta_N}{\alpha} \left(\frac{b}{a}T^* - 1 \right), \quad M^* = 1 - \frac{\beta}{\gamma} + \frac{\beta}{\gamma T^*}$$

Thus, the conditions for biological feasibility are:

- $S^* \geq 0$ iff $T^* \geq \frac{a}{b} = A$

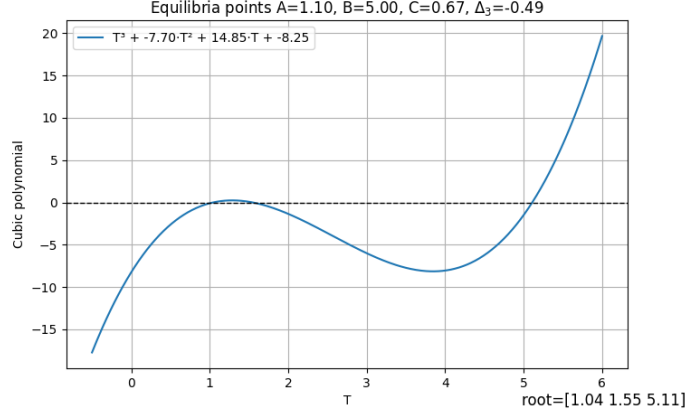


Figure 5: Cubic equation with three distinct real roots.

- $N^* \geq 0$ iff $T^* \geq 0$ (always true since we consider only positive roots)
- $M^* \geq 0$ iff $T^* \leq \frac{\beta}{\beta-\gamma} = \frac{B}{B-1} = 1 + \frac{1}{B-1}$ if $B = \frac{\beta}{\gamma} > 1$; always true if $B \leq 1$.

So the biological feasibility conditions can be summarized as follows:

$$T^* \geq A \quad (14)$$

$$T^* \leq 1 + \frac{1}{B-1} \quad \text{if } B > 1 \quad (15)$$

Let us consider again the particular case $A = a/b = 1$, if $B \leq 1$ there is only one equilibrium, the disease-free $(S^*, N^*, M^*, T^*) = (0, 1, 1, 1)$ otherwise if $B > 1$ the feasibility conditions (14)-(15) reduce to:

$$1 \leq T^* \leq 1 + \frac{1}{B-1}. \quad (16)$$

We can check easily that only the disease-free root $T^* = 1$ satisfies this condition, while the other two roots are outside this interval (see Figure 3). Indeed if $\Delta \geq 0$, we have $C \leq \frac{(B-1)^2}{4B}$, hence $\frac{B-1}{2C} \geq \frac{B-1}{2 \frac{(B-1)^2}{4B}} = \frac{2B}{B-1} \geq 2$ so the root $T^* = \frac{B-1+\sqrt{\Delta}}{2C} \geq 2$ is not feasible.

Moreover, the other root $T^* = \frac{B-1-\sqrt{\Delta}}{2C} \leq \frac{B-1}{2C}$. In order to be feasible, this root should satisfy $\frac{B-1-\sqrt{\Delta}}{2C} \leq \frac{B}{B-1}$, which implies $C \geq \frac{(B-1)^2}{2B}$, contradicting the assumption $\Delta \geq 0$.

In the general case $A \neq 1$, the feasibility conditions (14)-(15) have to be checked for each positive root found. For $A > 1$, we will show that none of the positive roots satisfy the feasibility conditions (see Figure 5) whereas for $A < 1$, there is always exactly one feasible root (in addition to the disease-free equilibrium) (see Figure 6). To this aim, we consider four cases depending on the signs of $A - 1$ and $B - 1$.

- Case $A > 1$ and $B \geq 1$.

Let us define $f(T^*)$ as the cubic polynomial in the left-hand side of equation (12). If $B > 1$ we have

$$\begin{aligned} f(T^*) &= T^{*3} - \frac{A}{C}(B-1)T^{*2} - AT^{*2} + \frac{A}{C}(B-1)T^* + \frac{A}{C}BT^* - \frac{AB}{C} \\ &= \frac{A}{C}(B-1)\left(T^* - \frac{B}{B-1}\right)(1 - T^*) + T^{*2}(T^* - A) \end{aligned}$$

So $f(T^*) > 0$, for $T^* \in [A, \frac{B}{B-1}]$ if $B > 1$.

If $B = 1$ then $f(T^*) = T^{*2}(T^* - A) + \frac{A}{C}(T^* - 1) > 0$ for $T^* > A$.

Hence, thanks to (14) and (15), there is no feasible root in this case.

- Case $A > 1$ and $B < 1$. Since

$$f(T^*) = \frac{A}{C}(B-1)(T^* - \frac{B}{B-1})(1 - T^*) + T^{*2}(T^* - A) > 0$$

for $T^* \geq A$ and $B < 1$, there is no feasible root in this case either.

- Case $A < 1$ and $B \geq 1$.

We compute $f(1) = 1 - A > 0$ and $f(A) = (A-1)\frac{A}{C}(A(1-B) + B) < 0$, so there is at least one feasible root in this case.

If $B > 1$, computing the derivative $f'(T^*)$ we have:

$$f'(T^*) = \frac{A}{C}(B-1)\left[\left(\frac{B}{B-1} - T^*\right) + (1 - T^*)\right] + 2T^*(T^* - A) + T^{*2} > 0$$

for $T^* \in [A, 1]$. So there is exactly one root in the interval $[A, 1]$. Since once again $f(T^*) = \frac{A}{C}(B-1)(T^* - \frac{B}{B-1})(1 - T^*) + T^{*2}(T^* - A)$ is positive for $T^* \in [1, \frac{B}{B-1}]$, there is exactly one feasible root in this case.

If $B = 1$, we have $f(T^*) = T^{*2}(T^* - A) + \frac{A}{C}(T^* - 1) > 0$ for $T^* > 1$ and $f'(T^*) = T^{*2} + 2T^*(T^* - A) + \frac{A}{C} > 0$ for $T^* \in [A, 1]$. So there is exactly one feasible root in this case too.

- Case $A < 1$ and $B < 1$.

Since $f(1) = 1 - A > 0$ and $f(A) = (A-1)\frac{A}{C}(A(1-B) + B) < 0$, there is at least one feasible root in this case too. Let $\tilde{T} = T^* - A$ and $\tilde{f}(\tilde{T}) = f(\tilde{T} + A)$. Then the equation $\tilde{f}(\tilde{T}) = 0$ is written as:

$$\begin{aligned} \tilde{T}^3 + \left(3A + \frac{A}{C}(1 - B - C)\right)\tilde{T}^2 + \left(3A^2 + 2\frac{A^2}{C}(1 - B - C) + \frac{A}{C}(2B - 1)\right)\tilde{T} \\ + \frac{A}{C}(A - 1)(A(1 - B) + B) \\ = \tilde{T}^3 + \left(2A + \frac{A}{C}(1 - B)\right)\tilde{T}^2 + \left(A^2 + 2\frac{A^2}{C}(1 - B) + \frac{A}{C}(2B - 1)\right)\tilde{T} \\ + \frac{A}{C}(A - 1)(A(1 - B) + B) = 0. \end{aligned}$$

Using *Descartes' rule of signs* we can see that there is exactly one positive root $\tilde{T}$ of the cubic equation $\tilde{f}(\tilde{T}) = 0$ as the coefficients of $\tilde{T}^3$ and $\tilde{T}^2$ are positive and the constant term is negative. So there is exactly one feasible root $T^* \geq A$ in this case too.

Let us summarize. The feasible equilibria points of the system can be classified as follows:

- Disease-free equilibrium: $(S^*, N^*, M^*, T^*) = (0, 1, 1, 1)$.
- Other equilibria with $S^* = 0$: $S^* = 0, N^* = 0, M^* = 0, T^* = R$, with arbitrary R ; $S^* = 0, N^* = 0, M^* = 1, T^* = 1$; $S^* = 0, N^* = 1, M^* = 0, T^* = R$, with arbitrary R .
- Feasible equilibria with $S^* \neq 0$: None when $A \geq 1$. When $A < 1$, there is only one biologically feasible equilibria with $S^* \neq 0$ (see proof and Figure 6).

Régions dans le cube $0 < A, B, C < 5$: 0 / 1 / 2 équilibres réalisables

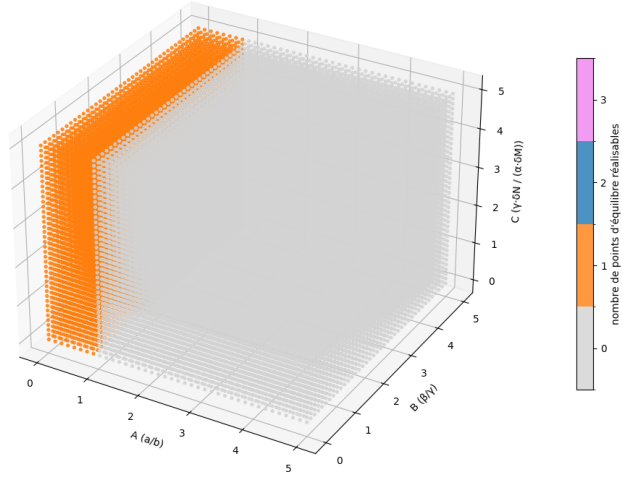


Figure 6: Feasible region in the (A, B, C) space.

2.3 Stability analysis

Stability analysis of the equilibria point (S^*, N^*, M^*, T^*) is done by linearizing the system around this point. The Jacobian matrix of the system is given by:

$$J = \begin{pmatrix} aN - bT & aS & 0 & -bS \\ \alpha N & \alpha S - \delta_N(2N - 1) & 0 & 0 \\ \beta & 0 & -\delta_M(2M - 1) & 0 \\ -\gamma T & 0 & \delta_M(1 - T) & -\gamma S - \delta_M M \end{pmatrix}$$

2.3.1 Stability of inflammation free equilibria ($S^* = 0$).

We analyze first the stability of the disease-free equilibrium point $(S^*, N^*, M^*, T^*) = (0, 1, 1, 1)$. At this point, the Jacobian matrix simplifies to:

$$J|_{(S^*, N^*, M^*, T^*)=(0,1,1,1)} = \begin{pmatrix} a - b & 0 & 0 & 0 \\ \alpha & -\delta_N & 0 & 0 \\ \beta & 0 & -\delta_M & 0 \\ -\gamma & 0 & 0 & -\delta_M \end{pmatrix}$$

We notice immediately that the eigenvalues of this matrix are $a - b$, $-\delta_N$ and $-\delta_M$. The last two are negative since $\delta_N, \delta_M > 0$. The remaining eigenvalue is $a - b$. Thus the disease-free equilibrium is stable if $a < b$ and unstable if $a > b$. This means that if the inflammatory effect of Neutrophils cells controlled by parameter a is less than the curing effect of the SL-TRM macrophages controlled by parameter b , the system will return to the disease-free state. Conversely, if $a > b$, the system may move away from this equilibrium, potentially leading to a chronic inflammatory state. The stability of other equilibria points can be analyzed similarly by evaluating the Jacobian matrix at those points and determining the eigenvalues.

For $S^* = 0, N^* = 0, M^* = 0, T^* = R$, with arbitrary R , the Jacobian matrix becomes:

$$J|_{(S^*, N^*, M^*, T^*)=(0,0,0,R)} = \begin{pmatrix} -bR & 0 & 0 & 0 \\ 0 & \delta_N & 0 & 0 \\ \beta & 0 & \delta_M & 0 \\ -\gamma R & 0 & \delta_M(1-R) & 0 \end{pmatrix}$$

The eigenvalues of this matrix are $-bR, \delta_N, \delta_M$, and 0. Since δ_N and δ_M are positive, this equilibrium point is unstable. Even if the immune signal will decrease exponentially fast, the populations of neutrophils and momac will grow out of bound. This unstable equilibrium will not be observed in practice.

For $S^* = 0, N^* = 0, M^* = 1, T^* = 1$, the Jacobian matrix is:

$$J|_{(S^*, N^*, M^*, T^*)=(0,0,1,1)} = \begin{pmatrix} -b & 0 & 0 & 0 \\ 0 & \delta_N & 0 & 0 \\ \beta & 0 & -\delta_M & 0 \\ -\gamma & 0 & 0 & -\delta_M \end{pmatrix}$$

The eigenvalues of this matrix are $-b, \delta_N, -\delta_M$, and $-\gamma$. Since there is a positive eigenvalue, this equilibrium point is unstable. Neutrophils can grow out of bound. This equilibrium will not be observed in practice neither.

For $S^* = 0, N^* = 1, M^* = 0, T^* = R$, with arbitrary R , the Jacobian matrix is

$$J|_{(S^*, N^*, M^*, T^*)=(0,1,0,R)} = \begin{pmatrix} a - bR & 0 & 0 & 0 \\ \alpha & -\delta_N & 0 & 0 \\ \beta & 0 & \delta_M & 0 \\ -\gamma R & 0 & \delta_M(1-R) & 0 \end{pmatrix}$$

There is a positive eigenvalue δ_M , hence this equilibrium point is unstable. Momac can grow out of bound. This equilibrium will not be observed in practice neither.

2.3.2 Stability of equilibrium with $S^* \neq 0$.

The stability of the equilibria with $S^* \neq 0$ can be analyzed similarly by evaluating the Jacobian matrix at those points and determining the eigenvalues. Taking into account the equilibrium equations (2), (3), and (5), the Jacobian matrix at (S^*, N^*, M^*, T^*) is:

$$J = \begin{pmatrix} 0 & aS & 0 & -bS \\ \alpha N & -\delta_N(N) & 0 & 0 \\ \beta & 0 & -\delta_M(2M-1) & 0 \\ -\gamma T & 0 & \delta_M(1-T) & -\delta_M \frac{M}{T} \end{pmatrix}$$

The eigenvalues of this matrix can be found by solving the characteristic polynomial $\det(J - \lambda I) = 0$. A straightforward computation shows that the characteristic polynomial is given by:

$$\begin{aligned} P(\lambda) = & \lambda^4 + \left(\delta_N N + \delta_M(2M-1) + \delta_M \frac{M}{T} \right) \lambda^3 \\ & + \left(\delta_N N \delta_M(2M-1 + \frac{M}{T}) + \delta_M^2(2M-1) \frac{M}{T} - aS\alpha N - bS\beta\gamma \right) \lambda^2 \\ & + \left(-aS\alpha N \delta_M(2M-1 + \frac{M}{T}) - bS\gamma T \delta_N N + \delta_N N \delta_M^2(2M-1) \frac{M}{T} - b\delta_M^2 M^2(1-T) \right) \lambda \\ & - aS\alpha N \delta_M^2(2M-1) \frac{M}{T} - b\delta_N N \delta_M^2 M^2(1-T) \end{aligned}$$

Due to the Descartes' rule of signs, we claim that the polynomial $P(\lambda)$ has one positive root (hence the equilibrium is unstable). Indeed, the coefficients of $P(\lambda)$ are positive for the terms in λ^4 and λ^3 , whereas the constant term is negative. To prove that there is only one change of sign and hence one positive root, we have to check that if the coefficient of λ is positive then the coefficient of λ^2 is positive too.

If the coefficient of λ is positive we have the following inequality:

$$aS\alpha N\delta_M(2M-1+\frac{M}{T})+b\delta_M^2M^2(1-T)+bS\gamma T\delta_N N<\delta_N N\delta_M^2(2M-1)\frac{M}{T}$$

Thanks to the positivity of all terms, we can deduce the following inequalities:

$$aS\alpha N\delta_M(2M-1+\frac{M}{T})<\delta_N N\delta_M^2(2M-1)\frac{M}{T}, \quad (17)$$

and

$$bS\gamma T<\delta_M^2(2M-1)\frac{M}{T}. \quad (18)$$

Inequality (17) implies that

$$aS\alpha N<aS\alpha N\left(\frac{\delta_M^2(2M-1+\frac{M}{T})^2}{\delta_M^2(2M-1)\frac{M}{T}}\right)<\delta_N N\delta_M(2M-1+\frac{M}{T}). \quad (19)$$

Due to (18) and (19), the coefficient of λ^2 is positive.

We proved that biologically feasible equilibrium with $S^* \neq 0$ only exists if $A < 1$ and is unstable. For $A \geq 1$ the only biologically feasible equilibrium is the disease-free equilibrium which is unstable since $a > b$.

2.4 Bifurcation diagram

We perform a bifurcation analysis with respect to the *non dimensional* parameters $A = a/b$. We restrict to the biologically relevant feasible equilibria. From the stability analysis above, we see that we have only one bifurcation at $A = 1$: for $A < 1$ there are two biologically feasible equilibria, one inflammatory one with $S^* \neq 0$ which is unstable and the disease-free equilibrium which is stable; for $A > 1$ the disease-free equilibrium is unstable and there is no other biologically feasible equilibrium. The number and stability of equilibria change as the parameter A crosses the critical value $A = 1$, indicating bifurcation points in the system's dynamics. We can plot the bifurcation diagram in the (A, S^*) plane (see Figure 7).

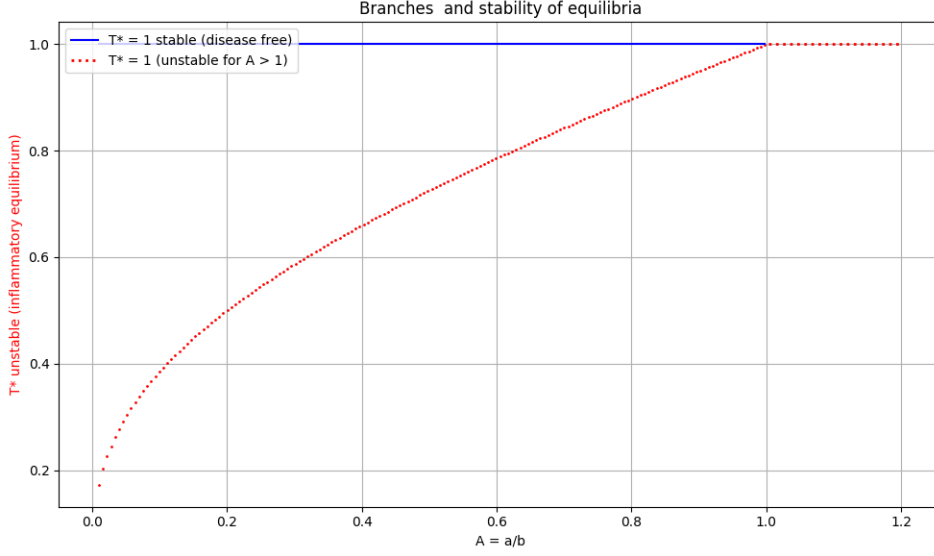


Figure 7: Bifurcation diagram in the (A, T^*) plane.

To Do List

- compute attracting sets for $A=1/$ when there is a zero eigenvalue (center manifold theory)
- is there hysteresis phenomenon? compute domain of persistence/ disease-free attraction basins in the (S, A) plane as in Torres et al. 2024
- Plot 4d phase space trajectories (projections)
- Plot 2d phase space trajectories (Mmac, Monocytes) or (neutrophiles, monocytes)(projections)
- Run numerical simulations with realistic rates
- perform sensitivity analysis of parameters
- calibrate results with immunology collaborators
- use optimal control theory to model treatments (corticosteroids, anti TNF alpha, ethanecerp, inhibitor of neutrophiles degranulation,...)

3 Numerical simulations

Here we reproduce some typical simulations of the model using the following parameters: $\alpha = 1, \beta = 1, \gamma = 1, \delta_N = 1, \delta_M = 1, a = 0.35, b = 1$, duration=20 days and initial conditions $S(0) = 1, N(0) = 1, M(0) = 1, T(0) = 1$. Now let us take $a = 0.38$, keeping the other parameters unchanged. We see clearly in fig. 8 that for $A = 0.35 < 1$ the system converges to the disease-free equilibrium, whereas for $A = 0.38 > 1$ the system diverges away from this equilibrium, indicating a chronic inflammatory state, see fig. 9.

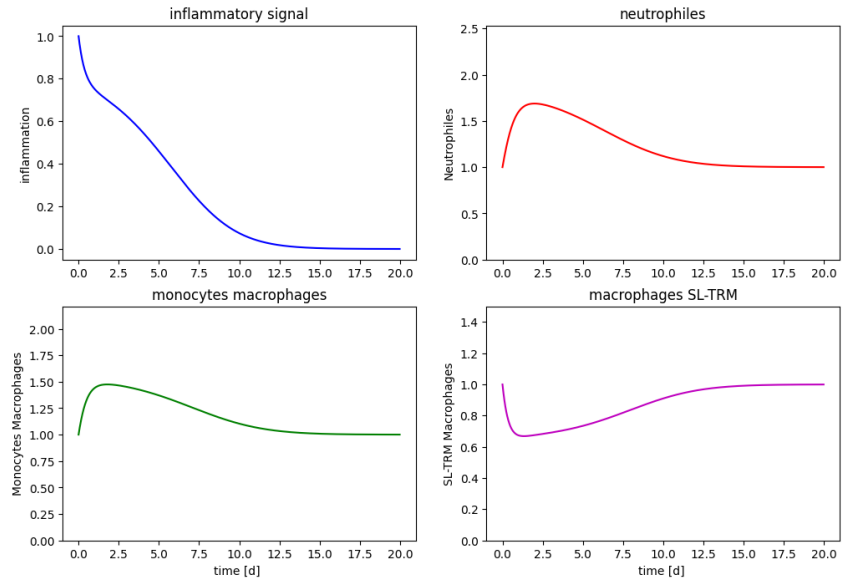


Figure 8: $A=0.35$

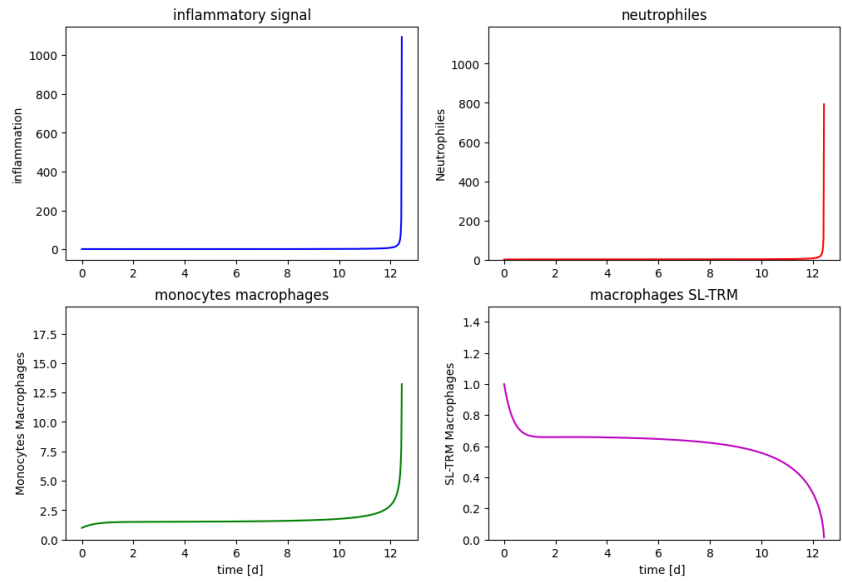


Figure 9: $A=0.38$

We can also decrease the initial immune signal $S(0)$ to see if the system converges to the disease-free equilibrium or diverges away from it. Halving the initial immune signal i.e. taking $S(0) = 0.5$ and $A = 0.38$, we obtain the following result: We see in fig. 10 that

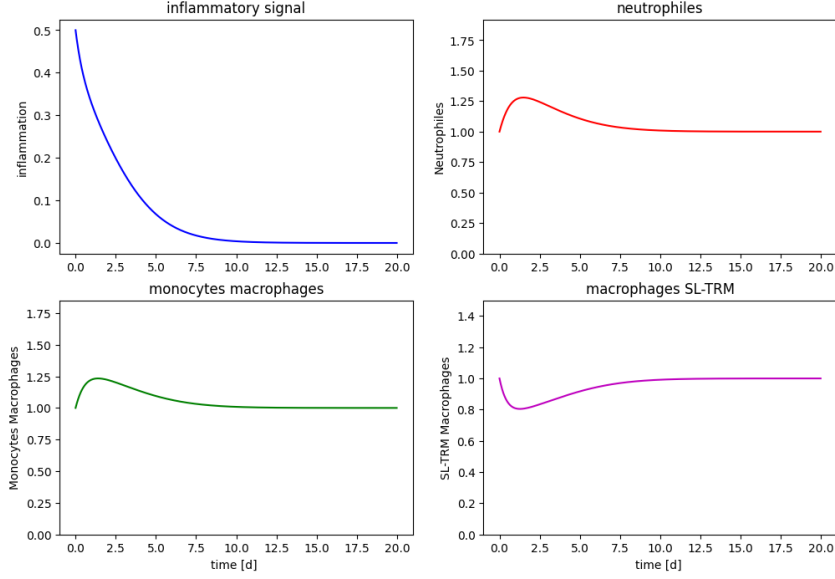


Figure 10: $A=0.38$, $S(0)=0.5$

even if $A = 0.38 > 1$, the system converges to the disease-free equilibrium, indicating that the initial immune signal is not strong enough to trigger a chronic inflammatory state. We can also start from a depressed immunity state, taking $S(0) = 0.5$, $N(0) = 1$, $M(0) = 0.1$, $T(0) = 0.05$ and $A = 0.38$. We observe in fig. 11 that the system diverges away from the disease-free equilibrium, indicating a chronic inflammatory state, even if the initial immune signal is low.

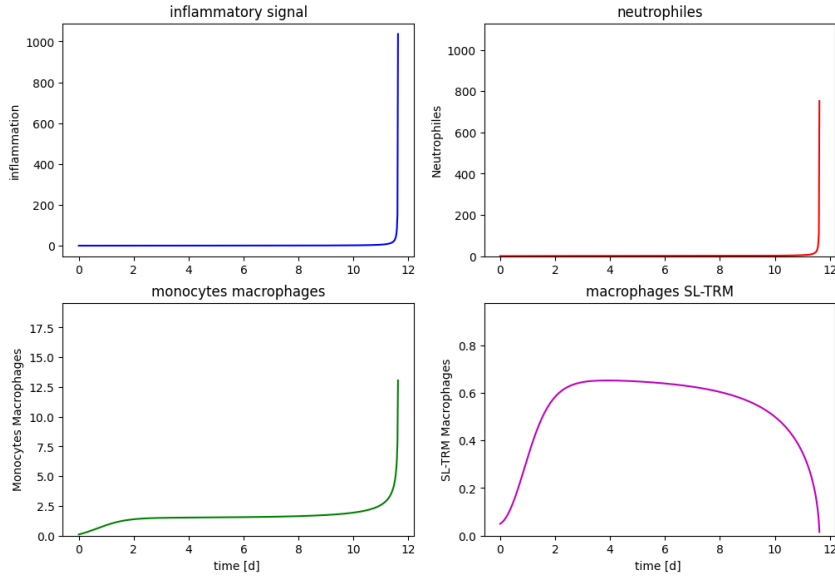


Figure 11: $A=0.38$, depressed initial conditions

To see how the parameter $A = a/b$ controls the healing prognosis we plot in Fig. 12 a (A, S) plane figure where for each value of A and starting value ($S(0) = S$, $N = 1$, $M =$

1, $T = 1$) we run the model for three weeks and see how far from the disease-free state the systems ends up. We color in green points where (S, M, N, T) is close to $(0, 1, 1, 1)$ up to a given tolerance (say 5%). We color in red points which end up far away from the equilibrium. We notice that for $A < 1$, the system converges to the disease-free equilibrium

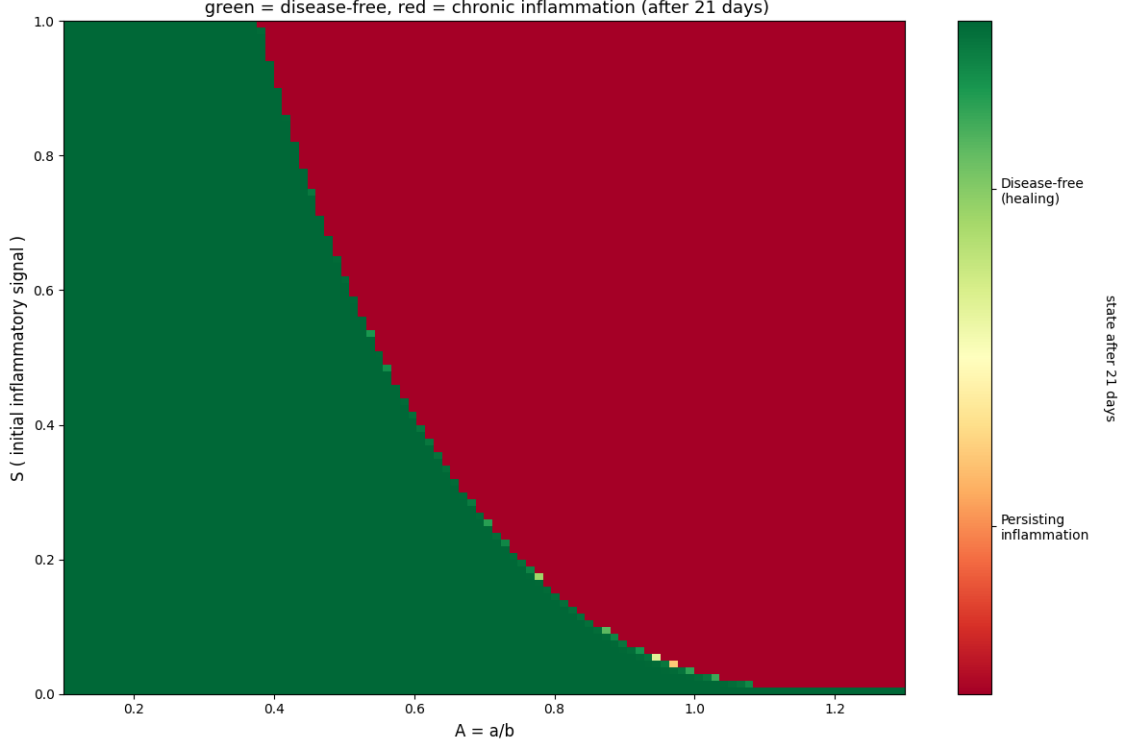


Figure 12: Healing w.r.t. $A = a/b$ and initial dose of immune signal S

provided that the initial immune signal $S(0)$ is below a given threshold. This threshold increases as A decreases, meaning that the system can tolerate higher initial immune signals without diverging away from the disease-free equilibrium when A is smaller. For $A > 1$, the system diverges away from the disease-free equilibrium for all values of the initial immune signal $S(0)$, indicating a chronic inflammatory state regardless of the initial conditions.

We can plot a phase space trajectory in the (S, N, M, T) 4D space, using color intensity to represent the S variable in fig. 13.

We can also plot some 2D projections of the 4D phase space trajectory using the tesseract representation in fig. 14.

We need to have realistic values for the parameters to feed the model.

The Python Jupyter notebook used to perform the simulations is available at <https://github.com/azerad/immunomath>.

Acknowledgments

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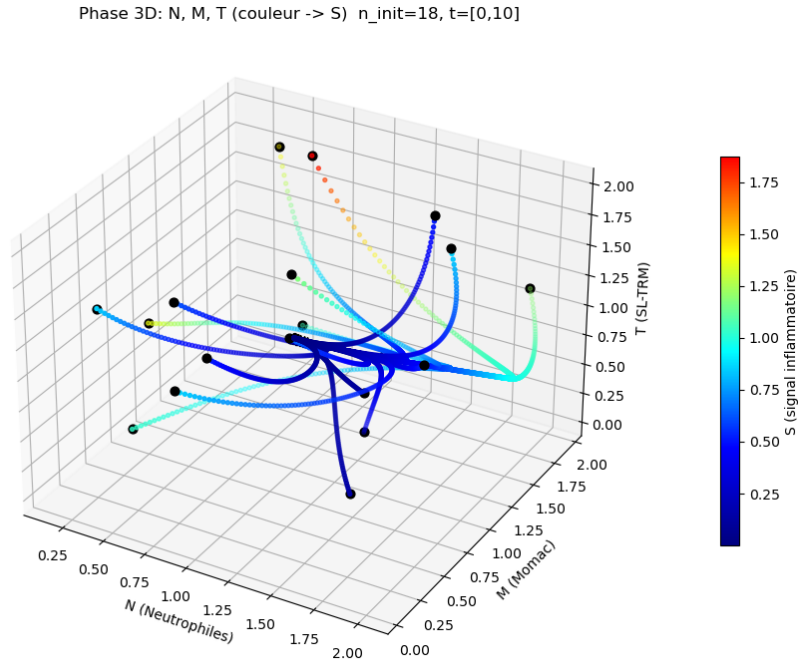


Figure 13: Phase space trajectory in the (S, N, M, T) space, S intensity corresponds to the color scale.

2D projection of 4D points with tesseract (hypercube) reference

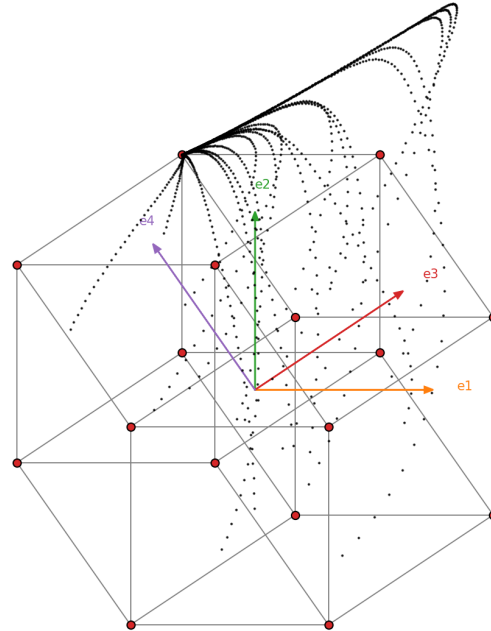


Figure 14: 2D projections of the 4D phase space trajectory using the tesseract representation.

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